DESIGN AND IMPLEMENTATION OF A MULTI-SENSOR SOIL MONITORING AND LORA COMMUNICATION SYSTEM FOR PRECISION FARMING APPLICATIONS

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Introduction

Agriculture plays a critical role in food security and livelihoods in Nigeria, particularly among smallholder farmers who depend heavily on the quality and productivity of their soils [1].
However, farmers face challenges in managing soil nutrients and water efficiently due to limited access to timely soil information [2][3].
Conventional farming practices often rely on manual assessment and routine application of water and fertilizer, which can result in resource wastage, increased costs, and reduced crop productivity [4].
Many existing precision-agriculture systems depend on internet connectivity, cellular networks, and grid electricity infrastructure that is frequently unreliable or unavailable in rural farming environments [5][6].

Research Goal

Aim: To design and implement a multi-sensor soil monitoring and LoRa communication system for precision farming.
Objectives:
To develop soil sensing nodes which will be able to sense multiple factors.
To develop a wireless communication module utilizing LoRa module which will transfer sensed data to a gateway.
To evaluate the performance of the developed system.

Methods

The system is designed as : Sensing and acquisition layer; wireless communication layer; edge processing and cloud layer; and visualization layer
ESP32-WROOM-32D field nodes (7-in-1 RS485 soil probe + NEO-6M GPS, solar powered) transmit over a 433 MHz LoRa link (SF9) every 3 minutes to an ESP32 gateway, which runs local advisories and syncs to a web dashboard.

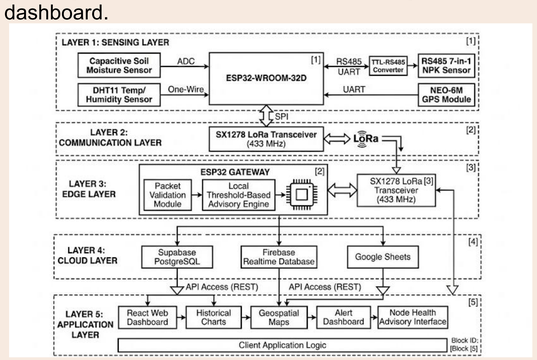


Figure 1: System architecture

Results



Figure 2: Field Deployment of the Solar-Powered Soil Monitoring Node

This project presents the design and field validation of a self-sustaining, multi-sensor soil monitoring and LoRa communication system for off-grid precision farming applications. Solar-powered ESP32-based field nodes integrate a 7-in-1 RS485 soil sensor for measuring soil moisture, temperature, pH, electrical conductivity (EC), nitrogen (N), phosphorus (P), and potassium (K), together with a NEO-6M GPS module for location tagging. The collected data are transmitted through a 433 MHz SX1278 LoRa communication link to a central gateway, eliminating dependence on cellular networks or Wi-Fi connectivity at the field level.

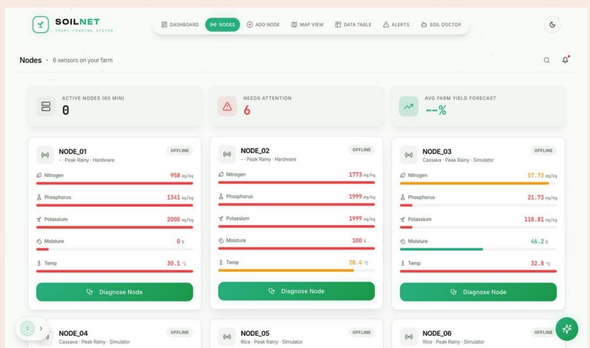


Figure 3: Soil Monitoring Dashboard

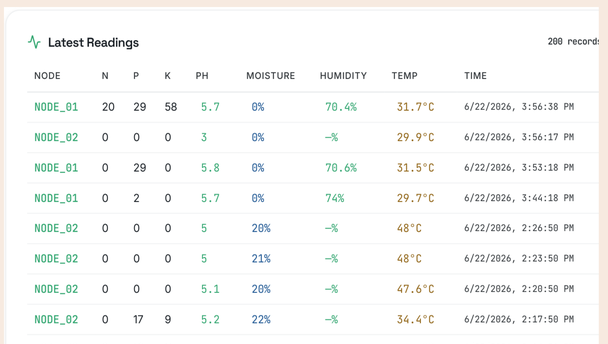


Figure 4: Sensor Data Logging Interface

The developed system successfully monitored and visualized **soil NPK, moisture, temperature, GPS location, and environmental data** from multiple sensor nodes. Recorded readings showed variations in soil conditions, including **N: 20.73–118.99 mg/kg, P: 17.45–39.98 mg/kg, K: 17.04–127.52 mg/kg, moisture: 0–40.3%, and temperature: 26.2–34.5°C**. The dashboard provided centralized data logging, node status monitoring, and automated alerts, demonstrating the system's suitability for precision farming and informed soil management.

Conclusion

This prototype shows that self-reliant edge-IoT sensing, LoRa telemetry and advisory generation can replace cloud-dependent precision-agriculture platforms in off-grid settings.
Field trials confirmed reliable multi-node telemetry, correctly differentiated advisories, and live dashboard tracking.

Acknowledgments

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